A layered pattern and the sharp form of Alon's conjecture on random universal permutations

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Abstract

The longest increasing subsequence of a random permutation of length n has length about $2\sqrt{n}$, so a random permutation contains the increasing pattern of length k with high probability when $n \geq (1+\varepsilon)k^2/4$ and not when $n \leq (1-\varepsilon)k^2/4$. Alon conjectured that, for every fixed $\varepsilon > 0$, a random permutation of length $(1+\varepsilon)k^2/4$ contains *every* pattern of length k with high probability. We give a proof that this fails for small ε: the layered pattern $2\ 1\ 4\ 3\ \cdots\ k\,(k-1)$ has a sharp containment threshold at $(1+\varepsilon_0)k^2/4$ for a constant $\varepsilon_0 > 0$. The proof combines an exact fibre formula comparing the blocks 12 and 21 in a direct sum, a planted copy of the pattern 3416725 that creates a defect fibre at a controlled cost, and the rate of convergence of a superadditive mean. The proof was found by an AI system and has not yet been verified by a specialist.

Declaration on the use of generative AI

The proofs in this note were found and written by Claude (Anthropic), working as a coding agent in the author's repository on 10 November 2026. The author does not claim credit for the proof ideas. At the time of writing the proofs have been re-derived by AI reviewers (separate sessions of the same system, without access to the writing session) and the deterministic lemmas have been tested by programs that these reviewers wrote without seeing each other's code; details are in the verification record at the end. The proofs have not yet been verified line by line by the author or read by a specialist. The note is circulated to obtain that verification.

1 Introduction

A permutation $\sigma \in S_n$ *contains* a pattern $\pi \in S_k$ if some k entries of σ are in the same relative order as π; it is *k-universal*, or a k-superpattern, if it contains every $\pi \in S_k$. Let σ_n be a uniform element of S_n. Since the longest increasing subsequence of σ_n has length $(2+o(1))\sqrt{n}$ with high probability[1] [1–3], σ_n is not k-universal with high probability when $n \leq (1-\varepsilon)k^2/4$.

Conjecture 1.1 (Alon; see [4, 5] and [6, Conjecture 1.1]). For every fixed $\varepsilon > 0$, a uniform random permutation of length $(1+\varepsilon)k^2/4$ is w.h.p. k-universal.

He and Kwan proved that length $2000\,k^2 \log\log k$ suffices [6]; a recent preprint of Chen and Wang announces length $32k^2$ [7]. The conjecture is motivated in [6] by analogy with random graphs, where cliques and independent sets are the hardest subgraphs to find: the monotone patterns are

[1] "With high probability" (w.h.p.) means with probability tending to one as $k \to \infty$.

believed to be the hardest patterns. He and Kwan also state the weaker conjecture that every *fixed* $\pi \in S_k$ is contained w.h.p. at length $(1+\varepsilon)k^2/4$ [6, Conjecture 5.3], and ask for the containment threshold of a fixed pattern.

We show that a layered pattern is strictly harder than the monotone ones. For a permutation σ let $\mathrm{LIS}(\sigma)$ be the length of its longest increasing subsequence and $L_{21}(\sigma)$ the largest m such that σ contains

$$21^{\oplus m} \;=\; 2\;1\;4\;3\;\cdots\;(2m)\,(2m-1),$$

the direct sum of m decreasing pairs.

Theorem 1.2. The limit $c_{21} = \lim_{n\to\infty} \mathbb{E}L_{21}(\sigma_n)/\sqrt{n}$ exists and $0 < c_{21} < 1$.

Corollary 1.3 (threshold for the layered pattern). Let $\varepsilon_0 = c_{21}^{-2} - 1 > 0$, let $\varepsilon > -1$ be fixed and $n = \lfloor (1+\varepsilon)k^2/4 \rfloor$. As $k \to \infty$, the probability that σ_n contains $21^{\oplus\lfloor k/2\rfloor}$ tends to 0 if $\varepsilon < \varepsilon_0$ and to 1 if $\varepsilon > \varepsilon_0$. In particular, for $0 < \varepsilon < \varepsilon_0$, σ_n is w.h.p. not k-universal.

So Conjecture 1.1 fails for $\varepsilon < \varepsilon_0$, and so does [6, Conjecture 5.3]. The main step is a lower bound for the gap between the two patterns in a Poissonised model.

Theorem 1.4 (deficit). Let P_T be a Poisson process of unit intensity on $[0,T]^2$. There is T_* such that for all $T \geq T_*$,

$$\mathbb{E}\,\mathrm{LIS}(P_T) - 2\,\mathbb{E}L_{21}(P_T) \;\geq\; 10^{-84}\,\frac{T}{(\ln T)^2}.$$

The constants in the proofs are explicit and very small; they carry no information. Simulations give $\mathbb{E}\lfloor \mathrm{LIS}(\sigma_n)/2 \rfloor - \mathbb{E}L_{21}(\sigma_n) = (0.00113 \pm 0.00002)\sqrt{n}$ for n from 10^3 to 10^8 (1.29 million permutations, both statistics computed on the same permutation; data and programs are in the repository cited at the end), that is $c_{21} \approx 0.99887$ and $\varepsilon_0 \approx 0.0023$. Together with [7], Corollary 1.3 would place the least constant C such that length Ck^2 is w.h.p. k-universal between $(1+\varepsilon_0)/4$ and 32.

1.1 Proof outline

Fibres (Section 2). Replacing a block 12 by 21 in a direct sum never increases the probability of containment. This is known: after reversal it is the case of chains of size two of the skew shape conjecture, proved by Jelínek and by Burstein and Pantone [8–10]. We need its proof, because it expresses the difference of the two probabilities as the expectation of a non-negative quantity over *fibres*, and identifies a three-point fibre, the *lens*, on which this quantity equals 1/3. Flipping blocks one at a time bounds $\mathbb{E}\,\mathrm{LIS} - 2\,\mathbb{E}L_{21}$ from below by a sum of non-negative *stage costs*.

The gadget (Sections 3 and 4). Seven points in the order 3416725, the pattern of [9], placed in the empty rectangle between two consecutive points of an optimal copy, create a lens. By the Mecke equation, the probability of finding such a planted configuration equals an expectation, over configurations with an empty rectangle, of a power of the area of that rectangle. So a stage is expensive unless the relevant step of the optimal copy spans a very small area.

Counting (Sections 5 and 6). First-moment bounds show that, with high probability, in every window of $90 \ln T$ consecutive steps of every optimal copy most steps span an area that is not small. Placing each flipped block at a random position within such a window makes one optimal copy serve all positions of the window, and gives Theorem 1.4.

Rate of convergence (Section 7). The mean of L_{21} in the Poisson square is superadditive and within $O(\sqrt{T \ln T})$ of its limit $c_{21}T$, as for longest common subsequences [11]. A deficit of order $T/(\ln T)^2$ is larger than this error, so it is a deficit of the constant.

1.2 Notation

Points of the plane are written $z = (z^{\mathsf{x}}, z^{\mathsf{y}})$; $z < z'$ means that both coordinates of z are strictly smaller, and $z \le z'$ that both are at most as large. $R(z, z') = \{w : z < w < z'\}$ is the open rectangle and $|R|$ its area. I_b is the increasing pattern of length b, $\oplus$ the direct sum, and "$Q \supseteq \gamma$" means that the point set Q contains the pattern γ. $\mathbb{E}[X; A]$ is the expectation of $X\mathbf{1}_A$, and e is the base of the natural logarithm (the letter e denotes a point). Statements are numbered within sections.

2 Fibres and the block-flip theorem

2.1 Setting

A *configuration* is a finite set $P \subset \mathbb{R}^2$ whose points have distinct abscissae and distinct ordinates. For a location z put $\mathrm{SW}(z) = \{r \in P : r < z\}$ and $\mathrm{NE}(z) = \{r \in P : r > z\}$. A set of points contains a pattern γ if some $|\gamma|$ of its points are order-isomorphic to γ; every set contains the empty pattern. A permutation σ is identified with the configuration $\{(i, \sigma(i))\}$, and a property of configurations is an *order-type invariant* if it depends only on the permutation that the configuration induces. A point a is *north-west* of c if $a^{\mathsf{x}} < c^{\mathsf{x}}$ and $a^{\mathsf{y}} > c^{\mathsf{y}}$; then $a \wedge c = (a^{\mathsf{x}}, c^{\mathsf{y}})$ and $a \vee c = (c^{\mathsf{x}}, a^{\mathsf{y}})$ are the *corners* of the pair.

P_T denotes a Poisson process of unit intensity on $[0, T]^2$. Almost surely it is a configuration, and given that it has n points its order type is uniform on S_n. Hence the expectation of an order-type invariant under P_T is a mixture over n of its expectations under σ_n.

Fix patterns α, β, possibly empty, and a configuration P. Let

$$\mathcal{U} = \mathcal{U}_\alpha(P) = \{z : \ \mathrm{SW}(z) \supseteq \alpha\}, \qquad \mathcal{V} = \mathcal{V}_\beta(P) = \{z : \ \mathrm{NE}(z) \supseteq \beta\}.$$

$\mathcal{U}$ is an up-set and $\mathcal{V}$ a down-set for $\le$. The *inside* points are $I = P \cap \mathcal{U} \cap \mathcal{V}$, the *outside* points are $O = P \setminus I$. Let X and Y be the sets of abscissae and of ordinates of the inside points. The *board* is the set of cells $(x, y) \in X \times Y$ that lie in $\mathcal{U}_\alpha(O) \cap \mathcal{V}_\beta(O)$. A *valid filling* is a bijection $\varphi : X \to Y$ whose graph Γ_φ lies in the board. A *D-inversion* of φ is a pair $a, c \in \Gamma_\varphi$ with a north-west of c, $a \wedge c \in \mathcal{U}_\alpha(O)$ and $a \vee c \in \mathcal{V}_\beta(O)$.

2.2 The theorem

Lemma 2.1 (reading the middle block). $P \supseteq \alpha \oplus 12 \oplus \beta$ if and only if there are $r < r'$ in P with $r \in \mathcal{U}$ and $r' \in \mathcal{V}$; then $r, r' \in I$. $P \supseteq \alpha \oplus 21 \oplus \beta$ if and only if there are $a, c \in P$, a north-west of c, with $a \wedge c \in \mathcal{U}$ and $a \vee c \in \mathcal{V}$; then $a, c \in I$.

Proof. A copy of $\alpha \oplus B \oplus \beta$, with B a block of two points, consists of a copy A of α, two middle points, and a copy A' of β, with A below-left of both middle points and A' above-right of both. For middle points $r < r'$ a suitable A exists iff $r \in \mathcal{U}$ and a suitable A' iff $r' \in \mathcal{V}$; then $r \in \mathcal{V}$ and $r' \in \mathcal{U}$ by monotonicity. For a north-west of c, "A below-left of both" means $A \subset \mathrm{SW}(a \wedge c)$ and "A' above-right of both" means $A' \subset \mathrm{NE}(a \vee c)$; and $a \wedge c \le a, c \le a \vee c$ gives $a, c \in \mathcal{U} \cap \mathcal{V}$. ∎

Lemma 2.2 (the regions are determined by the outside points). $\mathcal{U}_\alpha(P) = \mathcal{U}_\alpha(O)$ and $\mathcal{V}_\beta(P) = \mathcal{V}_\beta(O)$.

Proof. Let $z \in \mathcal{U}_\alpha(P)$ with α non-empty. Among the copies of α in $\mathrm{SW}(z)$ choose one, A, whose rightmost point is leftmost possible. If A contained an inside point r, then $r \in \mathcal{U}_\alpha(P)$ would give a copy of α in $\mathrm{SW}(r) \subset \mathrm{SW}(z)$ lying entirely to the left of r, contradicting the choice. So $A \subset O$ and $z \in \mathcal{U}_\alpha(O)$. The reverse inclusion is trivial, and the statement for $\mathcal{V}$ is symmetric. ■

Lemma 2.3 (fibres). Let φ be a valid filling of the board of P and $P_\varphi = O \cup \Gamma_\varphi$. Then the outside points of P_φ are exactly O, its inside points are exactly Γ_φ, and $\mathcal{U}_\alpha(P_\varphi) = \mathcal{U}_\alpha(O)$, $\mathcal{V}_\beta(P_\varphi) = \mathcal{V}_\beta(O)$. Consequently the configurations with outside set O and inside coordinate sets X, Y are exactly the P_φ, φ a valid filling.

Proof. Points of Γ_φ lie in $\mathcal{U}_\alpha(O) \cap \mathcal{V}_\beta(O) \subset \mathcal{U}_\alpha(P_\varphi) \cap \mathcal{V}_\beta(P_\varphi)$, so they are inside for P_φ. Let $o \in O$. By Lemma 2.2 for P, $o \notin \mathcal{U}_\alpha(O) \cap \mathcal{V}_\beta(O)$; say $o \notin \mathcal{U}_\alpha(O)$. If $o \in \mathcal{U}_\alpha(P_\varphi)$, take a copy A of α in $\mathrm{SW}(o)$ within P_φ. It is not contained in O, so it has a point $g \in \Gamma_\varphi$; but $g \in \mathcal{U}_\alpha(O)$ gives a copy of α in O below-left of $g < o$, a contradiction. So o is outside for P_φ, and the equalities of regions are Lemma 2.2 for P_φ. Conversely, if P' has outside set O and inside coordinate sets X, Y, its inside points lie in $\mathcal{U}_\alpha(P') \cap \mathcal{V}_\beta(P') = \mathcal{U}_\alpha(O) \cap \mathcal{V}_\beta(O)$ by Lemma 2.2 for P', hence on the board; so $P' = P_\varphi$ for a valid φ. ■

Lemma 2.4 (two sorting arguments). Suppose a board has a valid filling. (i) The decreasing bijection is valid, and it is the only valid filling without an increasing pair. (ii) There is a valid filling without a D-inversion.

Proof. (i) If a valid φ has an increasing pair $r < r'$, replace r, r' by $(r^{\mathsf{x}}, r'^{\mathsf{y}})$ and $(r'^{\mathsf{x}}, r^{\mathsf{y}})$. Both lie between $r \in \mathcal{U}_\alpha(O)$ and $r' \in \mathcal{V}_\beta(O)$ for $\leq$, hence on the board. The number of inversions increases; iterate. (ii) If a valid φ has a D-inversion (a, c), replace a, c by $a \wedge c$ and $a \vee c$. Then $a \wedge c \in \mathcal{U}_\alpha(O)$ by assumption and $a \wedge c \leq a \in \mathcal{V}_\beta(O)$; $a \vee c \in \mathcal{V}_\beta(O)$ by assumption and $a \vee c \geq a \in \mathcal{U}_\alpha(O)$. The number of inversions decreases; iterate. ■

For a configuration with $I \neq \emptyset$ let F be the number of valid fillings of its board and $F^\circ \geq 1$ the number of those without a D-inversion; put $\rho = (F^\circ - 1)/F$, and $\rho = 0$ if $I = \emptyset$. Both F and F° are order-type invariants.

Theorem 2.5 (block flip; fibre formula). For all patterns α, β and all n,

$$\Pr(\sigma_n \supseteq \alpha \oplus 12 \oplus \beta) - \Pr(\sigma_n \supseteq \alpha \oplus 21 \oplus \beta) \;=\; \mathbb{E}\,\rho(\sigma_n) \;\geq\; 0, \tag{2.1}$$

and the same identity holds with P_T in place of σ_n.

Proof. Partition S_n into classes according to (O, X, Y). By Lemma 2.3 a class with $I \neq \emptyset$ is in bijection with the valid fillings of its board, and $\mathcal{U}, \mathcal{V}$ are constant on it. By Lemmas 2.1 and 2.3, $P_\varphi \supseteq \alpha \oplus 12 \oplus \beta$ iff Γ_φ has an increasing pair, i.e. by Lemma 2.4(i) iff φ is not the decreasing bijection; and $P_\varphi \supseteq \alpha \oplus 21 \oplus \beta$ iff φ has a D-inversion. So a class with F members has $F - 1$ members containing the first pattern and $F - F^\circ$ containing the second, and $F^\circ \geq 1$ by Lemma 2.4(ii). Classes with $I = \emptyset$ contain neither. For P_T, average over the number of points; all quantities are order-type invariants. ■

After reversing permutations, the inequality in Theorem 2.5 is the case of chains of size two of the skew shape conjecture for transversal fillings, proved by Jelínek [8] and by Burstein and Pantone [9, Lemmas 1.4 and 1.5]; see [10, Theorem 2.5]. For α or β empty the two probabilities are equal [12].

2.3 Stage costs

For a set S of positive integers let $W_S = B_1 \oplus B_2 \oplus \cdots$ be the infinite pattern with $B_i = 21$ if $i \in S$ and $B_i = 12$ otherwise, let $W_S[1..t]$ be the pattern of its first t entries, and for a configuration P let

$$N_S(P) = \max\{t \geq 0 : \ P \supseteq W_S[1..t]\}.$$

Since $W_S[1..t]$ is contained in $W_S[1..t+1]$, $\mathbb{E}N_S(P) = \sum_{t\geq 1} \Pr(P \supseteq W_S[1..t])$ for a random configuration P. Clearly $N_\emptyset = \mathrm{LIS}$ and $2L_{21} \leq N_{\mathbb{N}}$. Note that $W_S[1..2m+1] = B_1 \oplus \cdots \oplus B_m \oplus 1$ whether or not $m+1 \in S$, because the first entry of a block is larger than everything before it.

Corollary 2.6 (monotonicity). Let P be σ_n or P_T. If $S \subset S'$ then $\Pr(P \supseteq W_{S'}[1..t]) \leq \Pr(P \supseteq W_S[1..t])$ for every t. Hence $\mathbb{E}N_{S'}(P) \leq \mathbb{E}N_S(P)$, and in particular

$$2\,\mathbb{E}L_{21}(P) \ \leq \ \mathbb{E}N_S(P) \ \leq \ \mathbb{E}\,\mathrm{LIS}(P) \qquad \text{for every } S.$$

Proof. $W_S[1..t]$ is obtained from $W_{S'}[1..t]$ by replacing complete blocks 21 by 12, one at a time; each replacement is an instance of Theorem 2.5. ■

For $i \in S$ let $\alpha_i = W_{S\cap[1,i-1]}[1..2i-2]$, the first $i-1$ blocks, and define the *cost of stage* i

$$\mathrm{Cost}_S(i) = \sum_{b\geq 0}\Big[\Pr(P_T \supseteq \alpha_i \oplus 12 \oplus I_b) - \Pr(P_T \supseteq \alpha_i \oplus 21 \oplus I_b)\Big] \ \geq 0.$$

It depends on S only through $S \cap [1, i-1]$.

Lemma 2.7 (telescoping). For every S,

$$\mathbb{E}\,\mathrm{LIS}(P_T) - 2\,\mathbb{E}L_{21}(P_T) \ \geq \ \mathbb{E}\,\mathrm{LIS}(P_T) - \mathbb{E}N_S(P_T) \ = \ \sum_{i\in S}\mathrm{Cost}_S(i).$$

Proof. The inequality is Corollary 2.6. For the identity fix t and let $i_1 < \cdots < i_r$ be the elements of S that are at most $t/2$; put $S^{(u)} = \{i_1, \ldots, i_u\}$ for $0 \leq u \leq r$. Then $W_S[1..t] = W_{S^{(r)}}[1..t]$, and

$$W_{S^{(u-1)}}[1..t] = \alpha_{i_u} \oplus 12 \oplus I_{t-2i_u}, \qquad W_{S^{(u)}}[1..t] = \alpha_{i_u} \oplus 21 \oplus I_{t-2i_u}.$$

So $\Pr(P_T \supseteq I_t) - \Pr(P_T \supseteq W_S[1..t])$ is a telescoping sum over u of non-negative terms. Summing over t and collecting, for each $i \in S$, the terms with $t = 2i + b$, $b \geq 0$, gives the claim. ■

Definition 2.8 (lens). A configuration P has a *lens at* (α, b) if, for the pair $(\alpha, \beta = I_b)$, there are exactly three inside points and, writing $x_1 < x_2 < x_3$ and $y_1 < y_2 < y_3$ for their abscissae and ordinates, the board consists of all nine cells except (x_1, y_1), which is not in $\mathcal{U}_\alpha(O)$, and (x_3, y_3), which is not in $\mathcal{V}_\beta(O)$.

For a point r of a configuration P let $h(r)$, the *height* of r, be the number of points of the longest increasing chain of P that starts at r. Then $\mathrm{NE}(r) \supseteq I_b$ iff $h(r) \geq b+1$, and a location z lies in $\mathcal{V}_{I_b}$, $b \geq 1$, iff some $r > z$ has $h(r) \geq b$.

Lemma 2.9 (lenses). Let α be a pattern.

(i) A configuration has a lens at (α, b) for at most one b.

(ii) $\displaystyle\sum_{b\geq 0}\Big[\Pr(P_T \supseteq \alpha\oplus 12\oplus I_b) - \Pr(P_T \supseteq \alpha\oplus 21\oplus I_b)\Big] \ \geq \ \frac{1}{3}\Pr(P_T \text{ has a lens at } (\alpha, b) \text{ for some } b).$

Proof. (i) A point r is inside for (α, I_b) iff $r \in \mathcal{U}_\alpha$ and $h(r) \geq b+1$, so the inside sets decrease as b increases. Suppose there is a lens at (α, b) and let r be its inside point with the largest abscissa. If $h(r) \geq b+2$, some $r' > r$ has $h(r') \geq b+1$, and $r' \in \mathcal{U}_\alpha$ because $r' > r$; then r' is an inside point to the right of r, which is impossible. So $h(r) = b+1$, and for $b' > b$ the inside set does not contain r and has at most two points.

(ii) On a lens board the valid fillings are 231, 312 and 321, so $F = 3$. In 321 the points in columns x_1, x_2 form a D-inversion, with corners (x_1, y_2) and (x_2, y_3). In 231 and in 312 each of the two inversions has (x_1, y_1) as lower corner or (x_3, y_3) as upper corner, so there is no D-inversion. Hence $F^\circ = 2$ and $\rho = 1/3$. Having a lens at (α, b) is an order-type invariant, so by (2.1) for P_T the bracket for b is at least $\frac{1}{3}\Pr(P_T \text{ has a lens at } (\alpha, b))$. By (i) these events are disjoint for different b. ■

3 The gadget

Throughout this section S is a finite set of positive integers, $\Pi = W_S$, and $t_0 = 2\max S$ ($t_0 = 0$ if $S = \emptyset$), so that Π is increasing after its first t_0 entries: $\Pi[1..t+1] = \Pi[1..t] \oplus 1$ for $t \geq t_0$. Fix a configuration P and let $N = N_S(P)$. For $t \geq 0$ let $\mathcal{U}_t = \mathcal{U}_{\Pi[1..t]}(P)$; these are up-sets, decreasing in t. The *level* of a point $r \in P$ is

$$\ell(r) = 1 + \max\{t : \ r \in \mathcal{U}_t\},$$

and $h(r)$ is its height, as defined before Lemma 2.9.

Lemma 3.1 (levels beyond t_0).

(i) For $t \geq t_0$: a location z lies in $\mathcal{U}_{t+1}$ iff some $r \in P$ with $r < z$ has $\ell(r) \geq t+1$.

(ii) If $\ell(r) \geq t_0 + 1$ then $\ell(r) = 1 + \max(\{t_0\} \cup \{\ell(r') : r' \in P,\ r' < r\})$. For any r: if $r' < r$ and $\ell(r') \geq t_0 + 1$ then $\ell(r) \geq \ell(r') + 1$. In particular points of the same level $\geq t_0 + 1$ are incomparable.

(iii) If $\ell(r) \geq t_0 + 1$ then $\ell(r) + h(r) - 1 \leq N$.

Proof. (i) A copy of $\Pi[1..t] \oplus 1$ in $\mathrm{SW}(z)$ is a copy A of $\Pi[1..t]$ and a point $r < z$ above-right of A; the latter says $r \in \mathcal{U}_t$, i.e. $\ell(r) \geq t+1$. (ii) Let $\bar{\ell}$ be the maximum of $\ell(r')$ over $r' < r$, with $\bar{\ell} = 0$ if there is no such point. If $\bar{\ell} \geq t_0 + 1$, (i) with $t = \bar{\ell} - 1$ gives $r \in \mathcal{U}_{\bar{\ell}}$ and (i) with $t = \bar{\ell}$ gives $r \notin \mathcal{U}_{\bar{\ell}+1}$. If $\bar{\ell} \leq t_0$, the hypothesis gives $r \in \mathcal{U}_{t_0}$ and (i) with $t = t_0$ gives $r \notin \mathcal{U}_{t_0+1}$. The second assertion is (i) with $t = \ell(r') - 1$. (iii) A copy of $\Pi[1..\ell(r)-1]$ below-left of r, the point r, and a chain of $h(r)-1$ points above r form a copy of $\Pi[1..\ell(r)+h(r)-1]$, because $\ell(r) - 1 \geq t_0$. ■

Anchors and outside conditions. Let $e < f$ be points of P and $R = R(e, f)$. A point $r \in P \setminus (R \cup \{e, f\})$ is called *lower* if $r < f$, *upper* if $r > e$, and *lateral* otherwise. No point is both lower and upper, and a lateral point is north-west or south-east of the whole rectangle. Consider the conditions:

- (O1) $\ell(e) \geq t_0 + 1$.
- (O2) $\ell(r) \leq \ell(e)$ for every lower point r.
- (O3) $h(r) \leq h(f)$ for every upper point r.
- (O4) No lateral point r has both $\ell(r) \geq \ell(e) + 2$ and $h(r) \geq h(f) + 2$.
- (O5) No lower point r with $\ell(r) = \ell(e)$ is north-west of e.
- (O6) No upper point r with $h(r) = h(f)$ is north-west of f.

Write $\mathcal{E}(e,f)$ for the conjunction of (O1)–(O6). Conditions (O5) and (O6) are tie-breaking rules, used only in Lemma 3.4.

Lemma 3.2 (the conditions do not see the rectangle). The numbers $\ell(e)$, $h(f)$ and the truth of $\mathcal{E}(e,f)$ are the same for P and for every configuration P' with $P' \setminus R = P \setminus R$.

Proof. If $r \notin R$ and ($r < f$ or $r = e$), then no point of R is below-left of r: such a point w would satisfy $e < w < r$, which forces $r \in R$ if $r < f$ and is absurd if $r = e$. So $\ell(r)$ depends only on $P \setminus R$. Symmetrically $h(r)$ depends only on $P \setminus R$ if $r \notin R$ and ($r > e$ or $r = f$). For a lateral point both $\mathrm{SW}(r)$ and $\mathrm{NE}(r)$ avoid R. The conditions involve only these quantities and the classification of points outside R. ■

The planted pattern. "$P \cap R \cong 3416725$" means that R contains exactly seven points of P, order-isomorphic to 3 4 1 6 7 2 5. From left to right we name them $u_1, q_1, u_2, q_2, v_1, q_3, v_2$. The comparable pairs among them are exactly

$$u_1 < q_1, q_2, v_1, v_2; \qquad u_2 < q_2, q_3, v_1, v_2; \qquad q_1 < q_2, v_1, v_2; \qquad q_2 < v_1; \qquad q_3 < v_2.$$

Lemma 3.3 (the gadget). Let $e < f$ in P satisfy (O1)–(O4) and suppose $P \cap R(e,f) \cong 3416725$. Put $p = \ell(e) + 1$ and $b = h(f) + 1$. Then P has a lens at $(\Pi[1..p], b)$, with inside points q_1, q_2, q_3. The levels and heights of the planted points are those of Table 1.

Table 1: The planted points. Rank means rank among the seven points.

Point	Abscissa rank	Ordinate rank	Level	Height	Inside for $(\Pi[1..p], I_b)$
u_1	1	3	p	$b+3$	no: level $< p+1$
q_1	2	4	$p+1$	$b+2$	yes
u_2	3	1	p	$b+2$	no: level $< p+1$
q_2	4	6	$p+2$	$b+1$	yes
v_1	5	7	$p+3$	b	no: height $< b+1$
q_3	6	2	$p+1$	$b+1$	yes
v_2	7	5	$p+2$	b	no: height $< b+1$

Proof. Let $Q = P \cap R$. A point of P below-left of a point of Q is in Q, equals e, or is lower; a point above-right of a point of Q is in Q, equals f, or is upper.

Levels. Every $r \in Q$ is above e, so $\ell(r) \geq \ell(e) + 1 \geq t_0 + 2$ and Lemma 3.1(ii) computes $\ell(r)$ from the points below-left of r. By (O2) those outside Q have level at most $\ell(e) = p - 1$, with equality for e. Going through the list of comparable pairs gives the column of levels.

Heights. By (O3) the points above-right of $r \in Q$ outside Q have height at most $h(f) = b - 1$, with equality for f. Going through the list downwards gives the column of heights.

Inside points. A point r is inside for $(\Pi[1..p], I_b)$ iff $r \in \mathcal{U}_p$ and $\mathrm{NE}(r) \supseteq I_b$, i.e. iff $\ell(r) \geq p+1$ and $h(r) \geq b+1$. In Q this holds exactly for q_1, q_2, q_3. It fails for e and for lower points, whose level is at most $p-1$; for f and for upper points, whose height is at most $b-1$; and for lateral points by (O4).

Board. The abscissae $x_1 < x_2 < x_3$ are those of q_1, q_2, q_3 and the ordinates $y_1 < y_2 < y_3$ those of q_3, q_1, q_2. By Lemma 2.2 cells may be tested in P: a cell is in $\mathcal{U}_p$ iff some point below-left of it has

level at least p (Lemma 3.1(i)), and in $\mathcal{V}_{I_b}$ iff some point above-right of it has height at least b. The points below-left of (x_1, y_1) are e and lower points — not u_1, whose ordinate lies between y_1 and y_2, and not u_2, which is to the right of x_1 — so they have level at most $p-1$ and $(x_1, y_1) \notin \mathcal{U}_p$. The cells $(x_1, y_2) > u_1$ and $(x_2, y_1) > u_2$ are in $\mathcal{U}_p$, and every other cell is $\geq$ one of them. Symmetrically, the points above-right of (x_3, y_3) are f and upper points — not v_1, which is to the left of x_3, and not v_2, whose ordinate lies between y_2 and y_3 — so $(x_3, y_3) \notin \mathcal{V}_{I_b}$; the cells $(x_2, y_3) < v_1$ and $(x_3, y_2) < v_2$ are in $\mathcal{V}_{I_b}$, and every other cell is $\leq$ one of them. ■

For $p \geq t_0 + 2$ let $\mathcal{G}_p(P)$ be the set of pairs $e < f$ of points of P with $\ell(e) = p-1$, $\mathcal{E}(e, f)$, and $P \cap R(e, f) \cong 3416725$.

Lemma 3.4 (at most one planted pair). $|\mathcal{G}_p(P)| \leq 1$, and if $\mathcal{G}_p(P) \neq \emptyset$ then P has a lens at $(\Pi[1..p], b)$ for some b.

Proof. The second assertion is Lemma 3.3. Let (e, f) and (e', f') be in $\mathcal{G}_p(P)$. By Lemma 3.3 each gives a lens at $(\Pi[1..p], b)$, with $b = h(f) + 1$, respectively $b = h(f') + 1$; by Lemma 2.9(i) the two values of b agree, so $h(f) = h(f')$, and the inside set $\{q_1, q_2, q_3\}$ is the same for both pairs and lies in both rectangles. The hypotheses are symmetric in the two pairs. Suppose $e \neq e'$. Both have level $p - 1 \geq t_0 + 1$, so they are incomparable by Lemma 3.1(ii), and we may assume that e' is north-west of e. Then $e' < q_1 < f$, and $e' \notin R(e, f) \cup \{e, f\}$ because e' is not above e; so e' is a lower point for (e, f), of the same level as e and north-west of e: this contradicts (O5) for (e, f). Suppose $f \neq f'$. They have the same height, so they are incomparable, and we may assume that f' is north-west of f. Then $f' > q_1 > e$, and $f' \notin R(e, f) \cup \{e, f\}$ because f' is not below f; so f' is an upper point for (e, f), of the same height as f and north-west of f: this contradicts (O6). ■

Lemma 3.5 (leftmost optimal copy). Suppose $N \geq t_0 + 2$. For $t_0 + 1 \leq i \leq N$ let $Z_i = \{r \in P : \ell(r) = i,\ h(r) = N - i + 1\}$ and let z_i be the point of Z_i with the smallest abscissa. Then every Z_i is non-empty, and for $t_0 + 1 \leq i < N$: $z_i < z_{i+1}$, $R(z_i, z_{i+1}) \cap P = \emptyset$, and $\mathcal{E}(z_i, z_{i+1})$ holds.

Proof. Take a copy of $\Pi[1..N]$ in P and let r_j be its point in position $j > t_0$. Then $\ell(r_j) \geq j$ and $h(r_j) \geq N - j + 1$, and Lemma 3.1(iii) forces equality, so $r_j \in Z_j$. Each Z_i is an antichain by Lemma 3.1(ii); in an antichain the leftmost point is the topmost, so z_i has the largest ordinate in Z_i.

Every $r \in Z_{i+1}$ has a predecessor in Z_i: since $\ell(r) = i + 1 \geq t_0 + 2$, the maximum in Lemma 3.1(ii) equals $i > t_0$ and is attained by a point $r' < r$ with $\ell(r') = i$; then $h(r') \geq h(r) + 1$, with equality by Lemma 3.1(iii). Every $r \in Z_i$ with $i < N$ has a successor in Z_{i+1}: some $r' > r$ has $h(r') = N - i$, and $\ell(r') \geq i + 1$ with equality by Lemma 3.1(iii). Let $w \in Z_i$ be a predecessor of z_{i+1} and $w' \in Z_{i+1}$ a successor of z_i. Then $z_i^{\mathsf{x}} \leq w^{\mathsf{x}} < z_{i+1}^{\mathsf{x}}$ and $z_i^{\mathsf{y}} < w'^{\mathsf{y}} \leq z_{i+1}^{\mathsf{y}}$, so $z_i < z_{i+1}$. A point of P strictly between them would have a level strictly between i and $i + 1$.

Write $e = z_i$, $f = z_{i+1}$; then $h(e) = N - i + 1$ and $h(f) = N - i$. (O1) is clear. (O2): if $r < f$ then either $\ell(r) \leq t_0 \leq i$, or $\ell(r) \geq t_0 + 1$ and $\ell(r) \leq \ell(f) - 1 = i$ by Lemma 3.1(ii). (O3): $r > e$ gives $h(r) \leq h(e) - 1 = h(f)$. (O4): $\ell(r) \geq i + 2$ and $h(r) \geq N - i + 2$ would contradict Lemma 3.1(iii). (O5): a lower point r with $\ell(r) = i$ has $h(r) \geq h(f) + 1 = N - i + 1$, with equality by Lemma 3.1(iii); so $r \in Z_i$, and it cannot be north-west of z_i. (O6): an upper point r with $h(r) = N - i$ has $\ell(r) \geq i + 1$, with equality; so $r \in Z_{i+1}$, and it cannot be north-west of z_{i+1}. ■

4 Resampling

Lemma 4.1 (resampling identity). Let S be finite, $\Pi = W_S$ and $t_0 = 2\max S$, with levels, heights and the conditions $\mathcal{E}$ as in Section 3, and let $p \geq t_0 + 2$. Then

$$\Pr(\mathcal{G}_p(P_T) \neq \emptyset) \;=\; \mathbb{E} \sum_{(e,f)} \mathbf{1}\{e < f,\ \ell(e) = p-1,\ \mathcal{E}(e,f),\ P_T \cap R(e,f) = \emptyset\}\, \frac{|R(e,f)|^7}{(7!)^2},$$

where the sum is over ordered pairs of distinct points of P_T.

Proof. By Lemma 3.4 the left side is $\mathbb{E}|\mathcal{G}_p(P_T)|$. By the bivariate Mecke equation [13, Theorem 4.4], for measurable $g \geq 0$,

$$\mathbb{E} \sum_{(e,f)} g(e, f, P_T \setminus \{e, f\}) = \iint \mathbb{E}\, g(e, f, P_T)\, \mathrm{d}e\, \mathrm{d}f,$$

the integral being over $([0,T]^2)^2$. By Lemma 3.2 the indicator of $\{\ell(e) = p-1,\ \mathcal{E}(e,f)\}$, evaluated in the configuration $\chi \cup \{e, f\}$, is a function $\Phi(e, f, \chi \setminus R(e,f))$; it is measurable, being an order-type invariant of finitely many points. Fix $e < f$ in $[0,T]^2$. The open rectangle $R = R(e,f)$ lies in $[0,T]^2$ and contains neither e nor f; $P_T \cap R$ and $P_T \setminus R$ are independent, and $P_T \cap R$ is a Poisson process on a set of area $A = |R|$. Seven independent uniform points have a prescribed order type with probability $1/7!$, so

$$\Pr(P_T \cap R \cong 3416725) = \exp(-A)\, \frac{A^7}{(7!)^2}, \qquad \Pr(P_T \cap R = \emptyset) = \exp(-A).$$

Apply the Mecke equation twice, with $g = \mathbf{1}\{e < f\}\, \Phi \cdot \mathbf{1}\{\chi \cap R \cong 3416725\}$ and with $g = \mathbf{1}\{e < f\}\, \Phi \cdot \mathbf{1}\{\chi \cap R = \emptyset\}\, A^7/(7!)^2$: both sides of the lemma equal $\iint_{e<f} \mathbb{E}\, \Phi(e, f, P_T \setminus R)\, \exp(-A) A^7/(7!)^2\, \mathrm{d}e\, \mathrm{d}f$. ∎

Proposition 4.2 (cost of a stage). Let S be finite, $\Pi = W_S$, $p \geq t_0 + 2$, $\alpha = \Pi[1..p]$, $N = N_S(P_T)$, and let (z_i) be the leftmost optimal copy of Lemma 3.5. Then

$$\sum_{b \geq 0} \Big[\Pr(P_T \supseteq \alpha \oplus 12 \oplus I_b) - \Pr(P_T \supseteq \alpha \oplus 21 \oplus I_b) \Big] \;\geq\; \frac{1}{3\,(7!)^2}\, \mathbb{E}\Big[\, |R(z_{p-1}, z_p)|^7;\ N \geq p \Big].$$

Proof. By Lemma 2.9(ii) and Lemma 3.4 the left side is at least $\frac{1}{3} \Pr(\mathcal{G}_p(P_T) \neq \emptyset)$. By Lemma 4.1 this is $\frac{1}{3}$ of an expected sum of non-negative terms, and on the event $N \geq p$ the pair (z_{p-1}, z_p) is one of them, by Lemma 3.5 with $i = p - 1 \geq t_0 + 1$. ∎

5 Three first-moment estimates

A *step* is a pair of points $z < z'$; it is *thin* if $|R(z,z')| < \theta$, where $\theta = 10^{-10}$. A chain $z_0 < \cdots < z_k$ has k steps, with increments $(\omega_i, \eta_i) = z_i - z_{i-1}$, and its *span* is $R(z_0, z_k)$.

Lemma 5.1 (a large rectangle contains a long chain). For every integer $k \geq 1$, the expected number of pairs $z < z'$ of points of P_T such that $|R(z,z')| \geq 4k^2$ and $P_T \cap R(z,z')$ contains no chain of k points is at most $T^4 \exp(-0.069\, k)$.

Proof. By the Mecke equation the expectation is the integral over $z < z'$ of the probability that the Poisson process in the open rectangle has no chain of k points. Cut a rectangle of area at least $4k^2$

into a $2k \times 2k$ grid of congruent cells, each of area at least 1. The $2k$ diagonal cells are non-empty independently with probability at least $1 - \exp(-1) > 0.632$, and points in distinct diagonal cells form a chain. By Hoeffding's inequality the probability that fewer than k of them are non-empty is at most $\exp(-2(0.132)^2 \cdot 2k) \le \exp(-0.069\,k)$. ■

Lemma 5.2 (chains of bounded span have few thin steps). For every integer $k \ge 1$ and $T \ge 2$, the expected number of chains $z_0 < \cdots < z_k$ in P_T with $|R(z_0, z_k)| \le 4k^2$ and at least $k/2$ thin steps is at most $(2\log_2 T + 5)\, T^2\, 3^{-k}$.

Proof. By the Mecke equation the expectation is the Lebesgue measure of the set of such $(z_0, \dots, z_k)$. Let $W = \sum \omega_i$, $H = \sum \eta_i$, $M = 2k$ and $R_0 = \lceil \log_2 T \rceil + 1$. Then $W, H \le T$ and $WH \le M^2$. There is an integer d with $|d| \le R_0$, $W \le M2^d$ and $H \le 2M2^{-d}$: if $W/M \ge 2^{-R_0}$ take $d = \lceil \log_2(W/M) \rceil$, so that $H \le M^2/W < 2M2^{-d}$ and $d \le \lceil \log_2(T/2) \rceil < R_0$; otherwise take $d = -R_0$ and use $H \le T \le 2^{R_0}$. Fix d and put $a = M2^d$, $c = 2M2^{-d}$, so $ac = 8k^2$. The indicator of $\{W \le a,\ H \le c\}$ is at most

$$\exp(k(1 - W/a) + k(1 - H/c)) = \exp(2k) \prod_i \exp(-k\omega_i/a - k\eta_i/c).$$

Choose which $\lceil k/2 \rceil$ steps are thin (at most 2^k ways), integrate z_0 over $[0, T]^2$ and each increment over $(0, \infty)^2$. A step without constraint contributes $ac/k^2 = 8$. A thin step contributes, after the substitution $\omega = (a/k)s$, $\eta = (c/k)s'$, the amount $8 \Pr(E_1 E_2 < \theta/8) \le 8 \cdot 2\sqrt{\theta/8} = 8\sqrt{\theta/2}$, where E_1, E_2 are independent standard exponential variables and we used $\Pr(E_1 E_2 < t) \le \Pr(\min(E_1, E_2) < \sqrt{t}) \le 2\sqrt{t}$. Since $8\sqrt{\theta/2} < 1$, the total for one d is at most $T^2\,(16\exp(2)(\theta/2)^{1/4})^k$, and $16\exp(2)(5 \cdot 10^{-11})^{1/4} < 0.315 < 1/3$. There are $2R_0 + 1 \le 2\log_2 T + 5$ values of d. ■

Lemma 5.3. For $T \ge 50$, $\Pr(2L_{21}(P_T) < 0.06\,T) \le \exp(-10^{-4}(T - 2))$.

Proof. The $\lfloor T/2 \rfloor$ squares $[2j, 2j+2)^2$ on the diagonal contain an inversion independently, with probability at least $4\exp(-4) > 0.073$ (exactly two points, forming an inversion), and inversions in distinct squares form a copy of $21^{\oplus m}$. By Hoeffding's inequality the number of such squares is at least $0.063\lfloor T/2 \rfloor \ge 0.03\,T$ except with probability $\exp(-2 \cdot 10^{-4}\lfloor T/2 \rfloor)$. ■

6 Proof of Theorem 1.4

Let $\ln T \ge 10$, $K = \lceil 45 \ln T \rceil \le 45.1 \ln T$ and $J = \lfloor 0.015\,T/K \rfloor$. Let $\xi_1, \dots, \xi_J$ be independent and uniform on $\{1, \dots, K\}$, independent of P_T, and put $i_0 = 0$, $i_j = i_{j-1} + K + \xi_j$, $S = \{i_1, \dots, i_J\}$. Lemma 2.7 holds for every value of S, hence on average:

$$\mathbb{E}\,\mathrm{LIS}(P_T) - 2\,\mathbb{E}L_{21}(P_T) \;\ge\; \sum_{j=1}^{J} \mathbb{E}[\mathrm{Cost}_S(i_j)]. \tag{6.1}$$

Fix j and $i_1 < \cdots < i_{j-1}$. Let $\Pi = W_{\{i_1, \dots, i_{j-1}\}}$, so that $t_0 = 2i_{j-1}$ in the notation of Section 3; let N be the length of the longest prefix of Π contained in P_T, and (z_ι) its leftmost optimal copy. $\mathrm{Cost}_S(i_j)$ depends only on $i_1, \dots, i_j$. For each of the K possible values $i = i_{j-1} + K + \xi$ of i_j, the pattern α_i is $\Pi[1..2i-2]$, because the blocks between i_{j-1} and i are 12 in both. So the conditional expectation of $\mathrm{Cost}_S(i_j)$ given $i_1, \dots, i_{j-1}$ is $\frac{1}{K}\sum_i C(i)$, where $C(i)$ is the left side of Proposition 4.2 for $p = 2i - 2$. The indices are related as follows, with $\iota_0 = t_0 + 2K - 2$.

Table 2: Bookkeeping for one stage.

Quantity	Value	Range as $\xi = 1, \dots, K$
candidate block i	$i_{j-1} + K + \xi$	$i_{j-1} + K + 1, \dots, i_{j-1} + 2K$
prefix length $p = 2i - 2$	$t_0 + 2K + 2\xi - 2$	at least $t_0 + 2K \geq t_0 + 2$
step used: (z_{p-1}, z_p)	$p - 1 = \iota_0 + 2\xi - 1$	steps number $2, 4, \dots, 2K$ of the window

Here the *window* is the chain $z_{\iota_0} < z_{\iota_0+1} < \cdots < z_{\iota_0+2K}$, whose $2K$ steps are numbered $1, \dots, 2K$. By Proposition 4.2,

$$C(i) \;\geq\; \frac{1}{3\,(7!)^2}\,\mathbb{E}\Big[\,|R(z_{p-1}, z_p)|^7;\; N \geq p\Big], \qquad p = 2i - 2.$$

Let $\mathcal{A}$ be the event that

- (A1) $N \geq \iota_0 + 2K$;
- (A2) no pair of points of P_T is as in Lemma 5.1 with $k = 2K$;
- (A3) no chain of P_T is as in Lemma 5.2 with $k = 2K$.

On $\mathcal{A}$ the window exists (Lemma 3.5; note $\iota_0 \geq t_0 + 1$) and $N \geq p$ for every candidate. A chain of P_T inside the span of the window has at most $2K - 1$ points, because levels increase strictly along a chain from z_{ι_0} to z_{ι_0+2K} (Lemma 3.1(ii)), from ι_0 to $\iota_0 + 2K$. By (A2) the span therefore has area less than $4(2K)^2$, and then by (A3) fewer than K of the $2K$ steps are thin. So at least $K + 1$ steps are not thin, and one of them has an even number. Hence

$$\sum_i C(i) \;\geq\; \frac{\theta^7}{3\,(7!)^2}\,\Pr(\mathcal{A}).$$

Probability of $\mathcal{A}$. (A1): $\iota_0 + 2K = 2i_{j-1} + 4K - 2 < 4KJ \leq 0.06\,T$, since $i_{j-1} \leq 2K(j-1)$; and $\Pr(N \geq t) = \Pr(P_T \supseteq \Pi[1..t]) \geq \Pr(P_T \supseteq W_{\mathbb{N}}[1..t]) \geq \Pr(2L_{21}(P_T) \geq t)$, by Corollary 2.6 and because a copy of $21^{\oplus\lceil t/2\rceil}$ contains $W_{\mathbb{N}}[1..t]$; so (A1) fails with probability at most $\exp(-10^{-4}(T-2))$ by Lemma 5.3. (A2) fails with probability at most $T^4 \exp(-0.069 \cdot 90 \ln T) \leq T^{-2}$. (A3) fails with probability at most $(2\log_2 T + 5)\,T^2\,3^{-90\ln T} \leq T^{-2}$. Hence $\Pr(\mathcal{A}) \geq 1/2$ for large T, and every stage satisfies

$$\mathbb{E}[\mathrm{Cost}_S(i_j) \,|\, i_1, \dots, i_{j-1}] \;\geq\; \frac{\theta^7}{6K\,(7!)^2}.$$

Summing over $j \leq J$ in (6.1), with $J/K \geq 7\cdot 10^{-6}\,T/(\ln T)^2$ for large T and $\theta^7/(6\,(7!)^2) > 6.5\cdot 10^{-79}$, gives Theorem 1.4. ■

Remarks. (i) With a second planted pattern, 1 4 5 2 7 8 3 6, which produces a lens one level higher, the odd-numbered steps of the window can be used as well and the bound improves to $c\,T/\ln T$; this variant is written out in the repository and is not used here. (ii) The logarithms come from (A2) and (A3), which are union bounds over all of $[0, T]^2$ and need windows of length of order $\ln T$. (iii) The randomisation of i_j is what allows one optimal copy, that of Π, to serve all K candidate positions.

7 From the deficit to the constant

Let P be a unit Poisson process on the quadrant $[0,\infty)^2$; for a rectangle R write $L_{21}(R)$ and $\mathrm{LIS}(R)$ for the functionals of $P \cap R$. Put

$$\Lambda_{21}(T) = \mathbb{E}L_{21}([0,T]^2), \qquad \Lambda(T) = \mathbb{E}\,\mathrm{LIS}([0,T]^2),$$

and let ν be the number of points of P in $[0,T]^2$. Let Y_n be L_{21} of n independent uniform points of a rectangle, which has the law of $L_{21}(\sigma_n)$, and $\mu_n = \mathbb{E}Y_n$; thus $\Lambda_{21}(T) = \mathbb{E}\mu_\nu$.

Standing facts.

- (F1) The law of $L_{21}(R)$ depends only on $|R|$ and is stochastically non-decreasing in it, because the maps $(x,y) \mapsto (sx, y/s)$ and translations preserve both the Poisson process and the order $<$.
- (F2) L_{21} changes by at most one when a point is added, removed or moved, and for an integer s the event $L_{21} \geq s$ is certified by $2s$ points. Also $L_{21} \leq \mathrm{LIS}$ pointwise, by taking one point from each block.
- (F3) Λ_{21} is non-decreasing and superadditive, $\Lambda_{21}(T_1+T_2) \geq \Lambda_{21}(T_1) + \Lambda_{21}(T_2)$, since copies in $[0,T_1]^2$ and $[T_1, T_1+T_2]^2$ concatenate. It is finite, being at most Λ, and $\Lambda(T) \leq 3T+1$ by the first-moment bound $\Pr(\mathrm{LIS}([0,T]^2) \geq k) \leq (\mathrm{e}T/k)^{2k}$; so the supremum below is finite. By Fekete's lemma, $c_{21} := \lim \Lambda_{21}(T)/T = \sup_T \Lambda_{21}(T)/T$, so $\Lambda_{21}(T) \leq c_{21}T$; and $c_{21} > 0$ by Lemma 5.3. In the same way $\Lambda(T) \leq T \cdot \lim \Lambda(T)/T = 2T$: here $\Lambda(T) = \mathbb{E}\,g(\nu)$ with $g(n) = \mathbb{E}\,\mathrm{LIS}(\sigma_n)$ non-decreasing, $g(n)/\sqrt{n} \to 2$ [1, 2], and $\nu/T^2 \to 1$ in L^1.
- (F4) $2\mu_n \leq \mathbb{E}\,\mathrm{LIS}(\sigma_n) \leq 3\sqrt{n}$ for $n \geq n_0$: the first inequality is Corollary 2.6 for σ_n; the second follows from $\Pr(\mathrm{LIS}(\sigma_n) \geq k) \leq \binom{n}{k}/k!$. Moreover $0 \leq \mu_{n+1} - \mu_n \leq 2\mu_{n+1}/(n+1) \leq 3/\sqrt{n+1}$: deleting a uniformly chosen point from $n+1$ points lowers L_{21} only if the point is one of the $2Y_{n+1}$ points of a fixed optimal copy.

Lemma 7.1 (concentration). Let T be large, $n_0 \leq n \leq 2T^2$, and let m_n be an integer median of Y_n.

(i) For $u \geq 0$: $\Pr(Y_n \geq m_n + u) \leq 2\exp(-u^2/(8(m_n+u)))$ and $\Pr(Y_n \leq m_n - u) \leq 2\exp(-u^2/(8m_n))$.

(ii) $m_n \leq 5T$ and $|m_n - \mu_n| \leq 17\sqrt{T}$. Consequently $\Pr(Y_n \geq \mu_n + 17\sqrt{T} + u) \leq 2\exp(-u^2/(80T))$ for $0 \leq u \leq 5T$, and $\Pr(Y_n \leq \mu_n - 17\sqrt{T} - u) \leq 2\exp(-u^2/(40T))$ for $u \geq 0$.

(iii) $|\Lambda_{21}(T) - \mu_{\lfloor T^2 \rfloor}| \leq 5\ln T + 1$, and $Y = L_{21}([0,T]^2)$ satisfies

$$\mathbb{E}(Y - \Lambda_{21}(T) - 18\sqrt{T} - 20\sqrt{T\ln T})^+ \;\leq\; T^{-4}.$$

Proof. (i) Y_n is a function of n independent points that is 1-Lipschitz and, by (F2), ψ-certifiable with $\psi(s) = 2s$. Talagrand's inequality [14] in the form [15, Theorem 7.7.1] gives $\Pr(Y_n \leq y - t\sqrt{2y})\,\Pr(Y_n \geq y) \leq \exp(-t^2/4)$ for integers $y \geq 1$ and real $t \geq 0$. We may assume $m_n \geq 1$, enlarging n_0. Take $y = m_n + \lceil u \rceil$ and $t = \lceil u \rceil/\sqrt{2y}$: the first factor is $\Pr(Y_n \leq m_n) \geq \frac{1}{2}$, and since Y_n is an integer and $v \mapsto v^2/(m_n+v)$ is increasing, the first bound follows. Take $y = m_n$ and $t = u/\sqrt{2m_n}$: the second factor is at least $\frac{1}{2}$, which gives the second bound.

(ii) By Markov's inequality and (F4), $m_n \leq 2\mu_n \leq 3\sqrt{n} \leq 5T$. By (i), $m_n - \mu_n \leq \mathbb{E}(m_n - Y_n)^+ = \int_0^\infty \Pr(Y_n \leq m_n - u)\,\mathrm{d}u \leq \sqrt{8\pi m_n} \leq 12\sqrt{T}$, and $\mu_n - m_n \leq \int_0^\infty \Pr(Y_n \geq m_n + u)\,\mathrm{d}u \leq$

$\int_0^{m_n} 2\exp(-u^2/(16m_n))\,\mathrm{d}u + \int_{m_n}^{\infty} 2\exp(-u/16)\,\mathrm{d}u \le \sqrt{16\pi m_n} + 32 \le 17\sqrt{T}$. The two tail bounds follow from (i), using $8(m_n+u) \le 80T$ for $u \le 5T$ and $8m_n \le 40T$.

(iii) Put $r = T\ln T$. By (F4), $|\mu_n - \mu_{\lfloor T^2\rfloor}| \le 5\ln T$ when $|n - T^2| \le r$. By a Chernoff bound for the Poisson distribution, $\Pr(|\nu - T^2| > r) \le 2\exp(-(\ln T)^2/3)$. Since $\mu_n \le n/2$, this gives $|\Lambda_{21}(T) - \mu_{\lfloor T^2\rfloor}| \le 5\ln T + \mathbb{E}[(\nu+T^2)/2;\, |\nu - T^2| > r] \le 5\ln T + 1$. So $\mu_n \le \Lambda_{21}(T) + \sqrt{T}$ when $|n - T^2| \le r$, and by (ii), for $0 \le u \le 5T$,

$$\Pr(Y \ge \Lambda_{21}(T) + 18\sqrt{T} + u) \;\le\; 2\exp(-u^2/(80T)) + 2\exp(-(\ln T)^2/3).$$

Let $s = 20\sqrt{T\ln T}$. The expectation in (iii) is $\int_s^{\infty} \Pr(Y \ge \Lambda_{21}(T) + 18\sqrt{T} + u)\,\mathrm{d}u$. The part over $[s, 5T]$ is at most $(80T/s)\,T^{-5} + 10T\exp(-(\ln T)^2/3)$. The part over $[5T, \infty)$ is at most $\mathbb{E}[Y; Y \ge 5T] \le \mathbb{E}[\nu; \mathrm{LIS}([0,T]^2) \ge 5T] \le (\mathbb{E}\nu^2\,(\mathrm{e}/5)^{10T})^{1/2}$, by (F2) and the first-moment bound $\Pr(\mathrm{LIS}([0,T]^2) \ge k) \le (\mathrm{e}T/k)^{2k}$. The sum is at most T^{-4} for large T. ■

Lemma 7.2 (rate of convergence). For large T,

$$\Lambda_{21}(2T) \le 2\,\Lambda_{21}(T) + 45\sqrt{T\ln T}, \qquad \text{and hence} \qquad c_{21}T - 200\sqrt{T\ln T} \;\le\; \Lambda_{21}(T) \;\le\; c_{21}T.$$

Proof. Let $m = L_{21}([0,2T]^2)$ and take a copy of $21^{\oplus m}$ with blocks $\beta_1, \dots, \beta_m$; let c_j be the upper-right corner of the bounding box of β_j. Both points of β_i are $\le c_j$ if $i \le j$ and $> c_j$ if $i > j$. Let j^* be the largest j such that the coordinates of c_j sum to at most $2T$, with $j^* = 0$ if there is none. If $j^* \ge 1$, the blocks $\beta_1, \dots, \beta_{j^*}$ lie in $R_x = [0,x] \times [0, 2T-x]$, where $x = c^{\times}_{j^*}$. If $j^* \le m-1$, the blocks $\beta_{j^*+2}, \dots, \beta_m$ lie in $R'_{x'} = [x', 2T] \times [2T - x', 2T]$, where $x' = c^{\times}_{j^*+1}$. Only β_{j^*+1} is not accounted for. Therefore, in all cases,

$$m \;\le\; 1 + M + M', \qquad M = \sup_{0 \le x \le 2T} L_{21}(R_x), \quad M' = \sup_{0 \le x \le 2T} L_{21}(R'_x).$$

The reflection $z \mapsto (2T, 2T) - z$ maps R_x to R'_{2T-x}, preserves the Poisson process on $[0,2T]^2$, and maps copies of $21^{\oplus m}$ to copies of $21^{\oplus m}$; so M' has the law of M and $\Lambda_{21}(2T) \le 1 + 2\,\mathbb{E}M$.

Put $\delta = 1/(2T)$. For $l = 0, \dots, \lceil 4T^2\rceil - 1$ and $l\delta \le x \le (l+1)\delta$, the rectangle R_x is contained in $\tilde{R}_l = [0, (l+1)\delta] \times [0, 2T - l\delta]$, a rectangle in the quadrant of area at most $T^2 + 1$. By (F1), $X_l = L_{21}(\tilde{R}_l)$ is stochastically dominated by $Y' = L_{21}([0,T']^2)$ with $T' = \sqrt{T^2+1}$; and $\Lambda_{21}(T') \le \Lambda_{21}(T) + 1$ by (F2), since $[0,T']^2 \setminus [0,T]^2$ has area 1. For every real a we have $M \le a + \sum_l (X_l - a)^+$, so $\mathbb{E}M \le a + \lceil 4T^2\rceil\,\mathbb{E}(Y' - a)^+$; no independence is used. With $a = \Lambda_{21}(T') + 18\sqrt{T'} + 20\sqrt{T'\ln T'}$, Lemma 7.1(iii) gives $\mathbb{E}M \le a + 1 \le \Lambda_{21}(T) + 22\sqrt{T\ln T}$ for large T. This proves the first inequality. Iterating it,

$$\frac{\Lambda_{21}(2^jT)}{2^jT} \;\le\; \frac{\Lambda_{21}(T)}{T} + \sum_{i<j} \frac{45\sqrt{2^iT\ln(2^iT)}}{2^{i+1}T} \;\le\; \frac{\Lambda_{21}(T)}{T} + 171\sqrt{\frac{\ln T}{T}},$$

because $\sqrt{\ln(2^iT)} \le \sqrt{\ln T}\,(1 + \sqrt{i})$ and $\frac{1}{2}\sum_{i\ge 0} 2^{-i/2}(1+\sqrt{i}) < 3.79$. Let $j \to \infty$ and use (F3). ■

Lemma 7.2 is the analogue for L_{21} of Alexander's theorem on the rate of convergence of the mean length of the longest common subsequence [11]; see also [16]. Cutting an optimal object along an antidiagonal and combining superadditivity with a concentration inequality is the scheme of [17].

Proof of Theorem 1.2. By Lemma 7.1(iii) with $T = \sqrt{n}$, $|\Lambda_{21}(\sqrt{n}) - \mu_n| \le 5\ln\sqrt{n} + 1$, so $\lim \mu_n/\sqrt{n} = c_{21}$, which is positive by (F3). For large T, Theorem 1.4, (F3) and Lemma 7.2

give

$$2c_{21}T - 400\sqrt{T\ln T} \;\le\; 2\,\Lambda_{21}(T) \;\le\; \Lambda(T) - 10^{-84}\frac{T}{(\ln T)^2} \;\le\; 2T - 10^{-84}\frac{T}{(\ln T)^2}.$$

Hence $1 - c_{21} \ge \frac{1}{2}\,10^{-84}(\ln T)^{-2} - 200\sqrt{\ln T/T}$, which is positive for large T. ■

Proof of Corollary 1.3. Let $m = \lfloor k/2 \rfloor$. Then $m/(c_{21}\sqrt{n}) \to (c_{21}\sqrt{1+\varepsilon})^{-1}$, which is larger than 1 if $\varepsilon < \varepsilon_0$ and smaller than 1 if $\varepsilon > \varepsilon_0$. By the previous proof, (F3) and Lemma 7.2, $c_{21}\sqrt{n} - O(n^{1/4}\sqrt{\ln n}) \le \mu_n \le c_{21}\sqrt{n} + O(\ln n)$. If $\varepsilon < \varepsilon_0$, there is $\kappa \in (0,1)$ with $m \ge \mu_n + 17n^{1/4} + \kappa\sqrt{n}$ for large k, and Lemma 7.1(ii) with $T = \sqrt{n}$ and $u = \kappa\sqrt{n}$ gives $\Pr(L_{21}(\sigma_n) \ge m) \le 2\exp(-\kappa^2\sqrt{n}/80) \to 0$. If $\varepsilon > \varepsilon_0$, there is $\kappa > 0$ with $m \le \mu_n - 17n^{1/4} - \kappa\sqrt{n}$ for large k, and the lower-tail bound of Lemma 7.1(ii) gives $\Pr(L_{21}(\sigma_n) < m) \to 0$. Finally $21^{\oplus\lfloor k/2 \rfloor}$ is contained in a pattern of length k, so a permutation that does not contain it is not k-universal. ■

8 Concluding remarks

The universality constant. Let C^* be the infimum of the constants C such that a uniform permutation of length Ck^2 is w.h.p. k-universal. Conjecture 1.1 states $C^* = 1/4$. Corollary 1.3 gives $C^* \ge (1+\varepsilon_0)/4$, and [7] announces $C^* \le 32$. We do not know the value of C^*, nor which patterns determine it.

Which pattern is hardest? Simulations of the analogous quantity for patterns $J_d \oplus J_d \oplus \cdots$ with decreasing layers of size d show deficits that grow with d: about $0.005\sqrt{n}$, $0.0075\sqrt{n}$ and $0.0105\sqrt{n}$ for $d = 3, 4, 6$, and up to $0.017\sqrt{n}$ for $d = 16$, against $0.0023\sqrt{n}$ for $d = 2$. A proof along the lines of this note would need the analogue of Theorem 2.5 for blocks of size d, which would follow from the skew shape conjecture [10, Conjecture 2.4] and is open for $d \ge 3$. Upper bounds for individual patterns are discussed in [18].

An effective bound. Theorem 1.2 gives no usable lower bound for $1 - c_{21}$, and the deficit proved at a finite scale is $T/(\ln T)^2$ rather than the order T suggested by simulation. The logarithms come from the union bounds (A2) and (A3). A deficit of order T with an explicit constant would follow from the same proof if, for some constants K and $a > 0$, a window of $2K$ consecutive steps of the leftmost optimal copy at a prescribed level contained an even-numbered step of area at least a with probability bounded below, uniformly in the stage. This is a statement about the environment seen from the starting point of a last-passage geodesic, and we expect it to require the theory of the Hammersley process rather than first-moment counting.

The monotone heuristic. The analogy with cliques in random graphs suggests that the most structured pattern is the hardest. For the comparison made here the relevant structure is local: every interior block 21 of a direct sum creates configurations, the lenses, in which two points can each continue an optimal copy but cannot both be used. The monotone pattern has no such blocks.

Verification record

This appendix records how the argument has been checked so far. It is not a substitute for refereeing.

Dependencies. Theorem 1.2 uses, from outside this note: the multivariate Mecke equation; Hoeffding's inequality; Talagrand's inequality for certifiable Lipschitz functions; Fekete's lemma; a Chernoff bound for the Poisson distribution; and $\mathbb{E}\,\mathrm{LIS}(\sigma_n)/\sqrt{n} \to 2$. Theorem 2.5 is proved in

full, although it is known. The logical order is: Theorem 2.5 → Corollary 2.6, Lemmas 2.7, 2.9 → Lemmas 3.1–3.5 → Lemma 4.1, Proposition 4.2 → Lemmas 5.1–5.3 → Theorem 1.4 → Lemmas 7.1, 7.2 → Theorem 1.2 and Corollary 1.3.

Where an error would most likely hide. In our judgement: Lemma 3.2 and its use in Lemma 4.1, that is, the claim that every outside condition is unaffected by the contents of the rectangle; Lemma 3.4, on which the passage from an expected number to a probability rests; the bookkeeping of indices in Section 6; and the passages between the Poisson model and uniform permutations.

Table 3: What has been checked, and how. Counts refer to brute-force computation from the definitions of Sections 2 and 3; different counts in one row come from programs written separately.

Statement	Check
Theorem 2.5	Exhaustive on S_n, $n \le 8$, for seven pairs (α, β), and $n \le 7$ for thirteen pairs by a second program
Theorem 2.5, in Lean 4	Exact counts over S_n, $n \le 9$, for six pairs (α, β), and the inequality for all α, β of size at most two and $n \le 8$, proved for the mathematical containment predicate; kernel-checked for $n = 7$ and for 1234 against 1324 at $n = 8$, compiled evaluation otherwise. The general theorem is not formalised
Lemma 2.7	Exact on S_7, S_8, S_9 for ten sets S
Lemmas 3.2, 3.3	Three programs: 7,756, 47,428 and 8,040 random plantings; negative controls fail when a hypothesis is dropped
Lemmas 2.9, 3.1	Tested on random configurations by one program
Lemma 3.4	4,730 random cases; 54,649 and 23,674 adversarial cases by two further programs; both tie-breaking rules are needed
Lemma 3.5	1,922, 6,714 and 30,888 consecutive pairs of leftmost optimal copies (three programs)
Lemma 4.1	Monte Carlo with a one-point plant (two programs); exact at leading order in T for the seven-point plant
Lemma 7.2	$\Lambda_{21}(2T) - 2\Lambda_{21}(T) \approx 0.65\, T^{1/3}$ for $T = 10, \dots, 400$; the cutting inequality holds in 3,300 of 3,300 samples
Proposition 4.2, Lemmas 5.1–5.3, 7.1	Written proofs; re-derived by the reviewers; all numerical constants recomputed
Theorems 1.2, 1.4, Corollary 1.3	Written proofs only; re-derived by four AI reviewers, two of which read only an earlier version of this note

Two of the reviews were of the working files in the repository and two of earlier versions of this note; the present text incorporates their corrections, and its last changes (notation and wording) have not been re-read. The constants of Theorems 1.2 and 1.4 are far outside the range of simulation, so these theorems cannot be tested numerically except through their consistency with the measured value $c_{21} \approx 0.99887$. Programs, raw outputs and the review reports are in the folders `experiments/w90-layered-deficit` and `experiments/w91-strict-deficit` of the repository https://github.com/adamhadani/superpatterns.

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